# ACE-LoRA: Graph-Attentive Context Enhancement for Parameter-Efficient Adaptation of Medical Vision-Language Models

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## Research Questions

- How can we bridge the specialization-generalization trade-off in medical vision-language models to capture fine-grained diagnostic cues while maintaining robust zero-shot generalization?
- How can hypergraph neural networks capture higher-order contextual interactions beyond pairwise attention to improve medical image-text alignment?
- How can we mitigate false negatives in medical contrastive learning where distinct samples may share identical clinical semantics?

## Core Idea

Medical VLMs face a fundamental trade-off: specialist models capture diagnostic details but generalize poorly, while generalist models retain broad semantics but lose fine-grained cues. ACE-LoRA addresses this by parameter-efficiently adapting generalist models using LoRA plus a novel hypergraph module (ACE-HGNN) that captures higher-order interactions among image patches and text tokens beyond simple pairwise attention. The key insight is that medical diagnosis requires understanding structured relationships among groups of visual/textual elements, not just global features. A label-guided InfoNCE loss prevents the model from pushing apart semantically similar medical samples during contrastive learning.

## Key Formula

**Hypergraph incidence matrix construction**

$$\boldsymbol{H}_{i,j} = \begin{cases} 1 & \text{if } i = j \\ \dfrac{\exp(\boldsymbol{S}_{i,j})}{\sum_{k \in \Omega_i} \exp(\boldsymbol{S}_{i,k})} & \text{if } j \in \Omega_i \text{ and } i \neq j \\ 0 & \text{otherwise} \end{cases}$$

H is the incidence matrix, S is the raw affinity matrix, Ω_i is the top-k neighbors for node i. This constructs hyperedges by selecting top-k similar tokens and applying softmax normalization while preserving self-connections. Critical for capturing higher-order token relationships in medical images.

**Label-guided InfoNCE loss**

$$\mathcal{L} = -\frac{1}{2B}\left( \sum_{i=1}^{B} \log \frac{\exp(\langle v_0^{'(i)}, t_0^{'(i)} \rangle / \tau)}{\sum_{k=1}^{B} 1_{i,k} \exp(\langle v_0^{'(i)}, t_0^{'(k)} \rangle / \tau)} + \sum_{i=1}^{B} \log \frac{\exp(\langle t_0^{'(i)}, v_0^{'(i)} \rangle / \tau)}{\sum_{k=1}^{B} 1_{i,k} \exp(\langle t_0^{'(i)}, v_0^{'(k)} \rangle / \tau)} \right)$$

v'_0 and t'_0 are refined global embeddings, τ is temperature, 𝟙_{i,k} is indicator function that equals 1 when disease labels differ or i=k, 0 otherwise. This prevents falsely pushing apart semantically similar medical samples during contrastive learning by excluding same-label pairs from negative samples.

## Key Figure

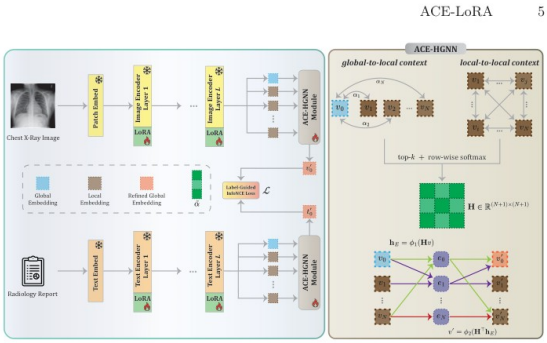


**Fig. 1: Overview of ACE-LoRA.** ACE-LoRA integrates low-rank adaptation modules into self-attention blocks of image and text encoders in a generalist medical VLM, and introduces ACE-HGNN, a hypergraph-based module that models high-order topological dependencies between local (e.g., image patches or report snippets) and global embeddings. For clarity, ACE-HGNN is described using image embeddings, though the same procedure is applied to text embeddings.

### 3.1 Preliminaries

The ViT-based [8] CLIP variant employs transformer encoders [46] for both image and text modalities. The image encoder processes $N$ non-overlapping patches with a prepended `[CLS]` token. For an input $z^{(l-1)} \in \mathbb{R}^{(N+1)\times d}$ at layer $l$, the update is formulated as:

$$z' = z^{(l-1)} + \mathrm{SA}(\mathrm{LN}(z^{(l-1)})), \quad z^{(l)} = z' + \mathrm{MLP}(\mathrm{LN}(z')), \tag{1}$$

where self-attention (SA) is defined by projection matrices $W_{\{Q,K,V,O\}} \in \mathbb{R}^{d\times d}$:

$$\mathrm{SA}(X) = \mathrm{softmax}\left(\frac{XW_Q W_K^T X^T}{\sqrt{d}}\right) XW_V W_O. \tag{2}$$

The final normalized global image and text embeddings are extracted from the last encoder layers. CLIP is trained to maximize the cosine similarity between matched image-text pairs in a shared latent space. Building on this foundation, BiomedCLIP [62] adapts the original CLIP for medical applications by training on the PMC-15M dataset with extended context length. *Nevertheless, it falls short in capturing domain-specific details within specialized subdomains, underscoring the need for effective fine-tuning to achieve robust domain adaptation.*

### 3.2 Integrating LoRA modules

Motivated by the success of LoRA [16] in VLMs for natural images [26, 59], we integrate LoRA modules into the query ($W_Q$), key ($W_K$), and value ($W_V$)

**Figure 1:** Shows the complete ACE-LoRA architecture with LoRA integration, ACE-HGNN hypergraph construction for local-global context modeling, and label-guided InfoNCE loss.

## Limitations

- Relies on CheXpert labeler for extracting disease labels, limiting applicability to domains without standardized label extraction tools
- ACE-HGNN module adds computational overhead during inference, though minimal compared to full fine-tuning
- Evaluated primarily on radiology and histopathology; generalization to other medical domains unclear

- Hypergraph construction uses fixed top-k selection which may not be optimal across all medical imaging modalities